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Assessment Tool - 2

CSA1613 - Data Warehousing and Data Mining

Q1) a) Define Data mining and KDD process.

A) Definition of Data mining :-

Data mining is the process of extracting useful, hidden and meaningful patterns or knowledge from large volumes of data using ~~static~~ statistical.

88/100

KDD :-

1) Data Selection
2) Data cleaning
3) Data Integration
4) Data Transformation

(b) Data Cleaning :-

Data Cleaning is the process of detecting and correcting inaccurate, incomplete, duplicate or inconsistent data.

=> Ex. Age 20, 22, -, 24

mean = (20+22+24) /3 = 22

missing value = 22

The Two common data Quality Issues

→ Missing values
→ noisy data

Q5 (a) Data Integration :-

Data integration is the process of combining data from multiple heterogeneous source into a single consistent data set.

-> Data redundancy.
-> Data inconsistency

* Handling Redundancy :-

-> Remove duplicate records
-> Use correlation analysis

(b) Covariance Analysis :-

Covariance analysis measures how two variable change together

*Formula :-

$$Cov(X,Y) = \frac{\Sigma (x_i - X)(y_i - Y)}{N}$$

Sample Covariance :-

$$Cov(X,Y) = \frac{\Sigma (x_i - X)(y_i - N)}{n-1}$$

Q3) (a) Stock Analysis:-

As Given:-

A	B
2	5
3	8
5	10
4	11
6	14

As Stock A increases, Stock B also generally increases.

=> The Convariance in ~~spa~~ positive, so ~~the~~ ~~the~~ two Stocks rise together.

(b)

Data Sources

↓

Data ware house

↓

Data cleaning & Integration

↓

Data base server

↓

Data mining Engine

↓

Pattern Evaluation module

↓

User / Decision maker.

Q4) a) Covariance

Given
$X = \{4, 5, 6, 7, 9\}$
$Y = \{6, 3, 7, 5, 6\}$

mean
$X = \frac{31}{5} = 6.2$

$Y = \frac{29}{5} = 5.8$

$=$ Sum $= -1.80$

$=$ Cov $= \frac{-1.80}{5} = -0.36$

Answer :- Covariance $= -0.36$

(b)

Major steps :-

Data cleaning

Data Integration

Data Transformation

Data Reduction

→ This sequence improve data quality removes inconsistence and prepare the data mining.

5) a) Equal - Frequency (equi-Depth) Binning.

Data:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40,

45, 46, 52, 70,

Total vals = 27

Bin 1 :-

13, 15, 16, 16, 19, 20, 20, 21, 22.

mean = 18

Smoothed values

18, 18, 18, 18, 18, 18, 18,

Bin 2:

22, 25, 25, 25, 30, 33, 33, 35

mean = 28

Smooth. vals.

of = 28, 28, 28, 28, 28, 28, 28, 28.

b) Data Discretization

Data discretization Converts continuous numerical values

into intervals (bins)

= Equal - width Binning

= Data 10 - 100

width = 30

Bins

-> 10-40

-> 40-70

-> 70-100

Advantage:

-> simple

-> Easy to implement.

Disadvantages:

-> sensitive to outliers.